02 The Crowding Room

Intraspecific competition & the logistic curve

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 32 min	**+ 28 min**	The Growth Room (01).
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/population-laboratories/` — then the Crowding tab

Individuals now get in each other's way: births fall and deaths rise as the population grows, and where the two meet the population stops changing. That point is the carrying capacity K. This guide asks what K does and does not mean, and where growth is fastest.

LEARNING OBJECTIVES

After the core path you should be able to

- Explain how density-dependent births and deaths produce a carrying capacity.
- Describe the logistic curve and show that K is approached from above as well as below.
- Locate the fastest population growth at $K/2$ and explain why.
- Explain how an Allee effect creates a threshold below which a population declines.

The extension adds

- Use $rK/4$ to predict the maximum total growth rate, and discuss maximum sustainable yield critically.
- Compare deterministic and stochastic versions of the same model.
- Explain why r sets the speed of the approach and K its height.

SUGGESTED ROUTES

- **35-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Fisheries or wildlife management session:** steps 1 and 6 with question 4.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 The S-shaped curve

8 min

SET UP

The defaults: N_0 = 10, r = 0.40, K = 400, time span 60, demographic noise off.

PREDICT FIRST

Where will the population level off? At what population size will it be growing fastest?

RUN

Press **Run**.

LOOK AT

the population curve against the purple K line and the pale exponential, and the *Total growth rate vs. density* hump.

EXPLAIN

Explain the gap between the logistic curve and the pale exponential.

2 Starting above K

8 min

SET UP

N_0 = **700**.

PREDICT FIRST

What will the population do?

RUN

Run.

LOOK AT

the *Births and deaths vs. density* panel at the start of the run.

EXPLAIN

Why is K called an attractor? What is happening to births and deaths above K?

3 The Allee effect

10 min

SET UP
Open *More parameters* and switch the **Allee effect** on, threshold A = 40. Run with N_0 = **30**, then N_0 = **50**.

PREDICT FIRST
What will each population do?

RUN
Run both.

EXPLAIN
Why is being rare itself dangerous here? Give two biological reasons an Allee effect might occur.

4 K or r?

6 min

SET UP
Allee off. *Compare two populations*: **On**, population 1 with K = 400 and population 2 with K = 600, same r and N_0.

PREDICT FIRST
Which population reaches half its own K first?

RUN
Run.

EXPLAIN
Which parameter sets how high the curve goes, and which sets how fast it climbs?

EXTENSION · MSC

5 Maximum sustainable yield

10 min

SET UP
One population, the defaults.

PREDICT FIRST
The total growth rate $rN(1 - N/K)$ peaks at $N = K/2$. Calculate the peak value.

RUN
Run and read the top of the hump on *Total growth rate vs. density*.

EXPLAIN
In the logistic model this is the largest harvest that could be taken forever. Give three reasons why setting a real fishery's quota at this value has repeatedly proved too much.

6 Noise near the ceiling

10 min

SET UP

Demographic noise: On, **20** replicates, K = **40**, N_0 = 10.

PREDICT FIRST

Will the replicates sit exactly at K? Could any go extinct?

RUN

Run twice; then raise K to 400 and run again.

EXPLAIN

How does the size of the fluctuations around K change with K, relative to K itself? What does that imply for small reserves?

7 Allee threshold with noise

8 min

SET UP

Noise on, 20 replicates, K = 400, Allee on with A = 40, N_0 = **45**.

PREDICT FIRST

Deterministically this population just escapes. With noise, what fraction of replicates will fall below A and be lost?

RUN

Run twice and count.

EXPLAIN

Why is a population near an unstable equilibrium especially at the mercy of chance?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc A deer population is held at 50% of K by hunting. Is that population growing fastest, slowest, or somewhere in between? Explain.
2. BSc Why does the room say K "belongs to the species" only in this model? What would make K change?
3. BSc Give an example of a species with an Allee effect, and explain the mechanism.
4. MSc The logistic model assumes density dependence is a straight line. How would a curved relationship move the peak of the growth hump?
5. MSc Why do theta-logistic and other generalisations exist? What does the logistic get wrong for large mammals?
6. MSc Explain why "maximum sustainable yield" is a property of the model rather than of the population.

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** The population levels off at $K = 400$ and grows fastest at 200, the peak of the hump.
- **Step 2.** It declines smoothly to K: above K deaths exceed births. K attracts from both sides.
- **Step 3.** From 30 the population declines to extinction; from 50 it grows to K. Mate finding, group defence or cooperative feeding can all cause an Allee effect.
- **Step 4.** Both reach half of their own K at the same time: r sets speed, K sets height.
- **Step 5.** $rK/4 = 0.4 \times 400/4 = 40$ individuals per unit time. Real stocks fluctuate, are age-structured, and are counted with error, and their density dependence need not be linear.
- **Step 6.** The replicates fluctuate around K; at $K = 40$ the swings are large relative to K and some can be lost. Relative fluctuations shrink as K grows.
- **Step 7.** A good share of replicates fall below 40 and are lost, though the deterministic run survives.